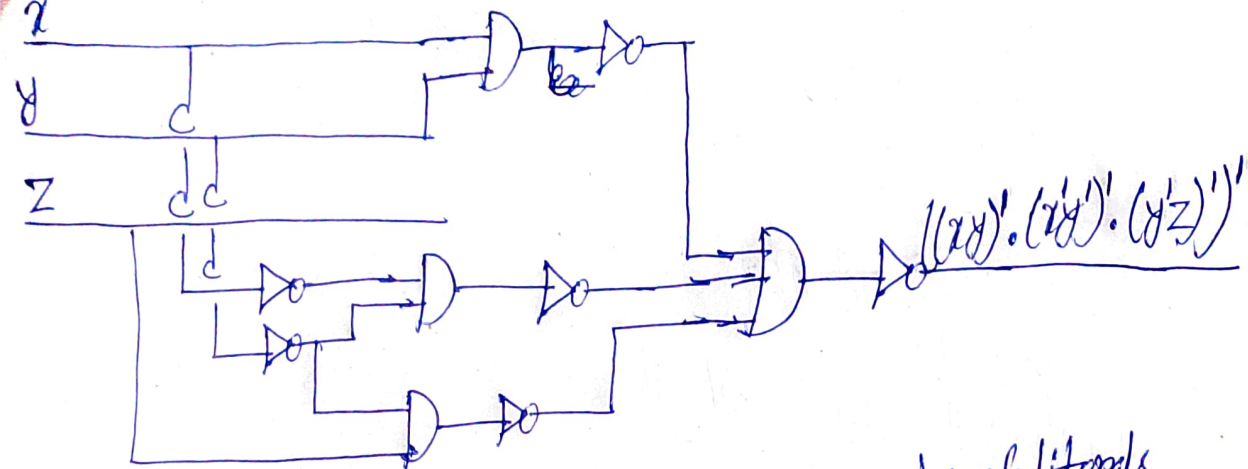




2-12: Simplify the functions T_1 and T_2 to a minimum number of literals.

A	B	C	T_1	T_2
0	0	0	1	0
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

$$\begin{aligned}
 T_1 &: A'B'C' + A'B'C + A'BC' \\
 &= (A'B'C' + A'B'C) + A'BC' \quad (\text{Theorem 1(a)}) \\
 &= A'B'(C' + C) + A'BC' \quad (\text{Theorem 4(a)}) \\
 &= A'B'(1) + A'BC' \quad (\text{Postulate 3(a)}) \\
 &= A'B' + A'BC' \quad (\text{Theorem 4(a)}) \\
 &= (A'B' + A'BC') + A'BC' \quad (\text{Theorem 4(b)}) \\
 &= (A'(B' + B)) + A'BC' \quad (\text{Theorem 4(b)}) \\
 &= (A'(1)) + A'BC' \quad (\text{Postulate 3(b)}) \\
 &= A' + A'BC' \quad (\text{Theorem 4(b)}) \\
 &= A'(1 + B) \quad (\text{Postulate 4(a)}) \\
 &= A'(1) \quad (\text{Postulate 5(a)}) \\
 &= A' \quad (\text{Postulate 2(b)}) \\
 &= A' \quad (\text{Postulate 4(a)})
 \end{aligned}$$

$$\begin{aligned}
 T_2 &: A'BC' + A'BC + AB'C' + ABC' + ABC \\
 &= A'BC' + A'BC + AB'C' + ABC' + (ABC' + ABC) \quad (\text{Theorem 1(a)})
 \end{aligned}$$